Exercise series 9: Concurrency Control

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Problem 1:

Consider the following classes of schedules: *conflict-serializable, serializable, recoverable,* and *avoids-cascading-rollback*. For each of the following *schedules*, state which of the preceding classes it belongs to.

T1	T2
Write(X)	
	Read(Y)
Read(Y)	
	Read(X)
	Commit
Commit	

T1	T2	T3
Read(X)		
	Read(Y)	
		Write(X)
	Read(X)	
	Commit	
		Commit
Read(Y)		
Commit		

T1	T2	T3
Read(X)		
	Write(X)	
	Commit	
Write(X)		
Commit		
		Write(X)
		Commit

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Problem 2:

Consider the following concurrence control protocols:
2PL, Strict 2PL, Rigorous 2PL, Timestamp with and without *Thomas Write Rule*
For each of the following schedules state which of these protocols allows it, that is allows the actions to occur in exactly the order shown.

T1	T2
Write(X)	
	Read(Y)
Read(Y)	
Commit	
	Read(X)
	Commit

T1	T2	T3
Read(X)		
	Read(Y)	
		Write(X)
	Read(X)	
	Commit	
		Commit
Read(Y)		
Commit		

T1	T2	T3
Read(X)		
	Write(X)	
	Commit	
Write(X)		
Commit		
		Write(X)
		Commit

Problem 3:

Answer the following questions:

1. Create a wait-for graph. Each transaction is a node. An arrow (or edge) is drawn between two nodes (or transaction), if transaction Ti has to wait-for transaction Tj to release a lock.
2. Does deadlock occur? How could you detect it?

T1	T2	T3	T4
S-lock(X)			
Read(X)			
	X-lock(Y)		
	Write(Y)		
S-lock(Y)			
		S-lock(W)	
		Read(W)	
	X-lock(W)		
			X-lock(Y)
		X-lock(X)	